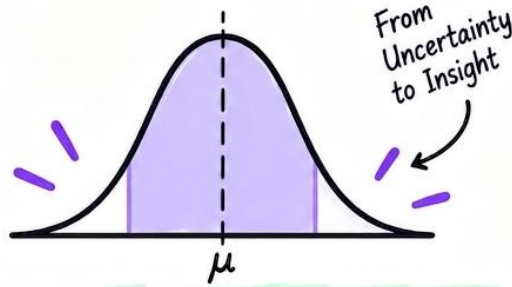


More  
Concepts.  
Deeper Understanding.  
Bigger Insights. 😊



Learn.  
Solve.  
Apply.  
Grow. 📊

# STATISTICS

## PART 2

- Concepts
  - Formulas
  - Examples
- Real-World Applications.

Same  
Statistics.  
Bigger Perspective.

— DataXOdyssey —

Better  
Questions.  
Stronger  
Decisions. 💡

# Probability Fundamentals

Probability helps us quantify how likely an event is to happen.  
It's all about uncertainty, expressed using numbers between 0 and 1.

Not  
certainty,  
but a way  
to measure  
possibility! 😊

## 1 What is Probability?

- Probability is the chance of an event happening.
- It is a number between 0 and 1.
- 0 means the event is impossible.
- 1 means the event is certain.
- Written as  $P(\text{Event})$  or  $P(A)$ .
- Useful in real life to make decisions under uncertainty (e.g. risk, forecasting, quality control, etc.).

## 2 Key Terms

- **Experiment:** A process with uncertain outcome (e.g. rolling a dice).
- **Outcome:** A possible result (e.g. 1, 2, 3, 4, 5, 6).
- **Sample Space (S):** Set of all possible outcomes.
- **Event (A):** A subset of the sample space (one or more outcomes).
- **Favourable Outcomes:** Outcomes that satisfy the event.

## 3 Basic Probability Formula

$$P(A) = \frac{\text{Number of favourable outcomes}}{\text{Total number of outcomes}}$$

- $0 \leq P(A) \leq 1$
- $P(A) = 0 \rightarrow$  impossible event
- $P(A) = 1 \rightarrow$  certain event



**Example:** Roll a fair dice.  
What is the probability of getting a 3?  
 $P(3) = \frac{1}{6}$  (1 favourable outcome out of 6)

## 4 Types of Events

- **Simple Event:** A single outcome (e.g. getting a 4).
- **Compound Event:** More than one outcome (e.g. getting an even number {2, 4, 6}).
- **Mutually Exclusive Events:** Cannot happen at the same time (e.g. getting a 2 and a 5).
- **Complementary Event:** The event not happening, denoted as  $A^c$ .

$$P(A^c) = 1 - P(A)$$

## 5 Addition Rule

For any two events A and B:

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

- If A and B are mutually exclusive ( $P(A \cap B) = 0$ ):  
 $P(A \cup B) = P(A) + P(B)$

**Example:**  
Roll a dice. What is the probability of getting a 2 or a 4?  
 $P(2 \text{ or } 4) = \frac{1}{6} + \frac{1}{6} = \frac{2}{6} = \frac{1}{3}$   
(both are mutually exclusive)

## 6 Multiplication Rule

For two dependent events:

$$P(A \cap B) = P(A) \times P(B | A)$$

For two independent events:

$$P(A \cap B) = P(A) \times P(B)$$

**Example (Independent):**

Toss two fair coins.  
What is the probability of getting two heads?  
 $P(H \text{ and } H) = \frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$



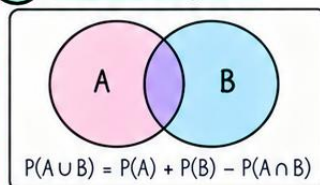
## 7 Conditional Probability

- Probability of an event A given that event B has already happened.

$$P(A | B) = \frac{P(A \cap B)}{P(B)}$$

- $P(A | B)$  is the updated probability based on new information.
- Used a lot in real-world problems (e.g. medical testing, spam detection).

## 8 Venn Diagram



$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

- Only A
- Only B
- Both A and B ( $A \cap B$ )

## 9 Key Takeaways

- ✓ Probability is always between 0 and 1.
- ✓ Use the basic formula for simple problems.
- ✓ Understand the difference between dependent and independent events.
- ✓ Apply addition and multiplication rules based on the situation.
- ✓ Conditional probability helps update probabilities with new information.
- ✓ Visual tools like Venn diagrams make it easier to understand.
- ✓ Probability is widely used in real life — from business to healthcare to AI!

"Probability doesn't predict the future, but it helps us be better prepared for it."

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# Events & Sample Space

Events and sample space are the building blocks of probability. They help us define what can happen and what we are interested in.

Define the possibilities, focus on the outcomes, that's probability!

## 1 What is Sample Space?

- Sample space (S) is the set of all possible outcomes of a random experiment.
- It includes every outcome that can occur.
- Usually denoted by S.

## 2 What is an Event?

- An event (E) is a subset of the sample space.
- It contains one or more outcomes.
- An event represents the outcome(s) we are interested in.
- Usually denoted by E.

## 3 Types of Events

- **Simple Event:** Contains only one outcome (e.g. getting a 3 on a die).
- **Compound Event:** Contains more than one outcome (e.g. getting an even number on a die).
- **Certain Event:** Contains all outcomes (e.g. getting a number from 1 to 6 on a die).
- **Impossible Event:** Contains no outcomes (e.g. getting a 7 on a standard die).

## 4 Example 1: Rolling a Die

Experiment: Roll a fair six-sided die.

Sample Space (S):

$S = \{1, 2, 3, 4, 5, 6\}$



Events:

- A = Getting an even number  
 $A = \{2, 4, 6\}$
- B = Getting a number  $> 4$   
 $B = \{5, 6\}$
- C = Getting a 3  
 $C = \{3\}$  (simple event)

## 5 Example 2: Tossing a Coin

Experiment: Toss a fair coin.

Sample Space (S):

$S = \{H, T\}$



Events:

- A = Getting a head  
 $A = \{H\}$  (simple event)
- B = Getting a tail  
 $B = \{T\}$  (simple event)
- C = Getting a head or tail  
 $C = \{H, T\}$  (certain event)
- D = Getting neither head nor tail  
 $D = \{\}$  (impossible event)

## 6 Example 3: Drawing a Card

Experiment: Draw one card from a standard deck of 52 cards.

Sample Space (S):

$S = \{52 \text{ cards}\}$

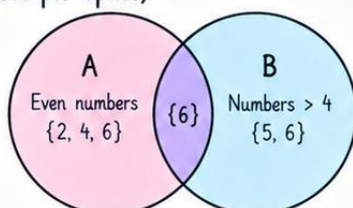


Events:

- A = Drawing an Ace  
 $A = \{A\spadesuit, A\heartsuit, A\clubsuit, A\diamondsuit\}$
- B = Drawing a King  
 $B = \{K\spadesuit, K\heartsuit, K\clubsuit, K\diamondsuit\}$
- C = Drawing a red card  
 $C = \{26 \text{ red cards}\}$
- D = Drawing a black card  
 $D = \{26 \text{ black cards}\}$

## 7 Venn Diagram View

S (Sample Space)



$S = \{1, 2, 3, 4, 5, 6\}$

A and B are events  
 $A \cap B = \{6\}$

## 8 Key Takeaways

- ✓ Sample space contains all possible outcomes.
- ✓ An event is a subset of the sample space.
- ✓ Events can be simple, compound, certain or impossible.
- ✓ Events help us focus on the outcomes we are interested in.
- ✓ Understanding events and sample space is essential for calculating probabilities.
- ✓ Always clearly define S and E before solving a problem.

"Probability starts with possibilities. Define the sample space, find the event, and the rest is just numbers!"

# Conditional Probability

Conditional probability tells us the probability of an event happening given that another event has **already** happened.

New information can change the probability!

## 1 What is Conditional Probability?

- Conditional probability is the probability of an event A occurring, given that event B has already occurred.
- Written as  $P(A | B)$  (read as "probability of A given B").
- It updates the probability of A based on the **new information** that B has happened.
- Only defined when  $P(B) > 0$ .

## 2 Formula

$$P(A | B) = \frac{P(A \cap B)}{P(B)}$$

- $P(A | B)$  = probability of A given B
- $P(A \cap B)$  = probability of both A and B occurring
- $P(B)$  = probability of B occurring (must be  $> 0$ )

## 3 Key Points

- ✓  $P(A | B)$  is not the same as  $P(A)$ .
- ✓  $P(A | B)$  can be different from  $P(A | C)$  if B and C are different events.
- ✓ If A and B are independent, then  $P(A | B) = P(A)$ .
- ✓ Conditional probability is widely used in real-world scenarios (e.g. medical testing, spam filters, risk analysis).

## 4 Example 1: Card Deck

A card is drawn from a standard deck of 52 cards. What is the probability of drawing a King given that the card is a face card?

**Solution:**

- Face cards = {J, Q, K} → 12 cards
- Kings among face cards = 4

$$P(\text{King} | \text{Face card}) = \frac{4}{12} = \frac{1}{3}$$

## 5 Example 2: Medical Test

A test for a disease is 95% accurate. If a person actually has the disease, what is the probability that the test is positive?

**Solution:**

- $P(\text{Test} + | \text{Disease}) = 0.95$

This is a conditional probability: probability of a positive test result given that the person has the disease.

## 6 Example 3: Using a Table

In a group of 100 students, 40 like Math (M), 30 like Science (S) and 20 like both.

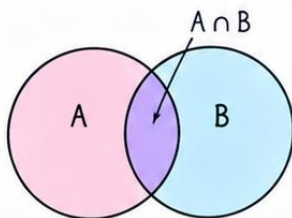
|               | Like Science | Not Science | Total |
|---------------|--------------|-------------|-------|
| Like Math (M) | 20           | 20          | 40    |
| Not Math      | 10           | 50          | 60    |
| Total         | 30           | 70          | 100   |

Find:  $P(M | S) = ?$

**Solution:**

$$P(M | S) = \frac{P(M \cap S)}{P(S)} = \frac{20}{30} = \frac{2}{3}$$

## 7 Venn Diagram View



$P(A | B)$  = Portion of B that is also in A

$P(B | A)$  = Portion of A that is also in B

## 8 Important Cases

- Independent Events:**  
If A and B are independent,  
 $P(A | B) = P(A)$
- Mutually Exclusive Events:**  
If A and B cannot happen together,  
 $P(A | B) = 0$
- $P(B) = 0$ :  
 $P(A | B)$  is undefined

## 9 Key Takeaways

- ✓ Conditional probability uses new information.
- ✓  $P(A | B) = P(A \cap B) / P(B)$ .
- ✓ It can change the probability of an event.
- ✓ If events are independent,  $P(A | B) = P(A)$ .
- ✓ Useful in real-world problems like healthcare, finance, recommendation systems, etc.

"Probability is not just about what can happen, but how the situation changes when we know more."

# Independent vs Dependent Events

Events can be **independent** (one doesn't affect the other) or **dependent** (one event changes the probability of the other).

Whether events are independent or dependent depends on the situation! 😊

## 1 What are Independent Events?

- Two events are independent if the occurrence of one event does not affect the probability of the other.
- The probability of the second event remains the same, no matter what happens with the first event.
- Formula:

$$P(A \cap B) = P(A) \times P(B)$$

Equivalently:

$$P(A | B) = P(A) \text{ and } P(B | A) = P(B)$$

## 2 What are Dependent Events?

- Two events are dependent if the occurrence of one event affects the probability of the other.
- The probability of the second event changes based on whether the first event has occurred.
- Formula:

$$P(A \cap B) = P(A) \times P(B | A)$$

(or)

$$P(A \cap B) = P(B) \times P(A | B)$$

## 3 Real-Life Examples

### Independent Events Example:

Tossing two coins.

A = First coin is Head (H)

B = Second coin is Head (H)

$$P(A) = \frac{1}{2}, P(B) = \frac{1}{2}$$

$$P(A \cap B) = \frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$$



The result of the first coin does not affect the second coin.

### Dependent Events Example:

Drawing two cards (without replacement).

A = First card is an Ace

B = Second card is an Ace

$$P(A) = \frac{4}{52}$$

$$P(B | A) = \frac{3}{51}$$

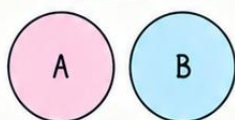
$$P(A \cap B) = \frac{4}{52} \times \frac{3}{51} = \frac{1}{221}$$



After drawing one Ace, the probability of the second Ace changes.

## 4 Visual Comparison

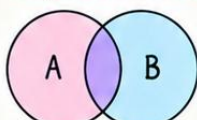
### Independent Events



- Events do not affect each other.
- $P(A \cap B) = P(A) \times P(B)$
- Knowing A has occurred does not change  $P(B)$ .

Example: Tossing two coins, rolling two dice.

### Dependent Events



- Events affect each other.
- $P(A \cap B) = P(A) \times P(B | A)$
- Knowing A has occurred changes  $P(B)$ .

Example: Drawing cards without replacement, choosing balls from a bag without replacement.

## 5 Key Differences

| Aspect                  | Independent Events                                           | Dependent Events                                                                 |
|-------------------------|--------------------------------------------------------------|----------------------------------------------------------------------------------|
| Definition              | One event does not affect the probability of the other.      | One event affects the probability of the other.                                  |
| Formula                 | $P(A \cap B) = P(A) \times P(B)$                             | $P(A \cap B) = P(A) \times P(B   A)$                                             |
| Conditional Probability | $P(A   B) = P(A)$<br>$P(B   A) = P(B)$                       | $P(A   B) \neq P(A)$<br>$P(B   A) \neq P(B)$                                     |
| Real-life Examples      | Coin tosses, rolling dice, random sampling with replacement. | Drawing cards without replacement, picking items from a bag without replacement. |

## 6 Quick Practice

Q1: A bag has 5 red and 5 blue balls. You draw one ball, note the colour, put it back and draw again. Are the two draws independent?

A) Yes B) No

Answer: A (with replacement)

Q2: A bag has 5 red and 5 blue balls. You draw one ball and do not put it back. Are the two draws independent?

A) Yes B) No

Answer: B (without replacement)

## 7 Key Takeaways

- ✓ Independent events: one event does not affect the other.
- ✓ Dependent events: one event changes the probability of the other.
- ✓ Use  $P(A \cap B) = P(A) \times P(B)$  for independent events.
- ✓ Use  $P(A \cap B) = P(A) \times P(B | A)$  for dependent events.
- ✓ Always check if the situation involves "with replacement" (independent) or "without replacement" (dependent).
- ✓ Real-world problems often involve dependent events.

"Same formula, different story – it all depends on whether the first event changes the second!"

# Bayes' Theorem

Bayes' Theorem helps us update our belief (probability) about an event based on new evidence (additional information).

It's not about being certain, but about updating what we know! 😊

## 1 What is Bayes' Theorem?

- Bayes' Theorem gives the probability of a cause (A) given an effect (B) has occurred.
- It uses prior knowledge and new evidence to update the probability.
- Very useful in real-life problems like medical testing, spam detection, fraud detection, etc.

## 2 Formula

$$P(A | B) = \frac{P(B | A) \times P(A)}{P(B)}$$

- $P(A | B)$  = Posterior probability (updated probability of A given B)
- $P(B | A)$  = Likelihood (probability of B given A)
- $P(A)$  = Prior probability (initial probability of A)
- $P(B)$  = Evidence (total probability of B)

## 3 Understanding the Terms

- Prior Probability ( $P(A)$ ):** Our initial belief before seeing the new evidence.
- Likelihood ( $P(B | A)$ ):** Probability of seeing the evidence if A is true.
- Evidence ( $P(B)$ ):** Total probability of seeing the evidence (can happen due to multiple causes).
- Posterior Probability ( $P(A | B)$ ):** Updated probability of A after seeing the evidence.

## 4 Example: Medical Test

**Problem:**

A disease affects 1% of the population. A test is 99% accurate (i.e. 99% chance of positive if you have the disease) and has a 5% false positive rate (i.e. 5% chance of positive even if you don't have the disease).

If a person tests positive, what is the probability that they actually have the disease?



## 5 Given Information

- $P(D) = 0.01$  (1% have the disease)
- $P(\neg D) = 0.99$  (99% do not have it)
- $P(+ | D) = 0.99$  (true positive rate)
- $P(+ | \neg D) = 0.05$  (false positive rate)

**We need to find:**

$P(D | +) = ?$   
(probability of having the disease given a positive test result)

## 6 Solution

Using Bayes' Theorem:

$$P(D | +) = \frac{P(+ | D) \times P(D)}{P(+)}$$

First, calculate  $P(+)$ :

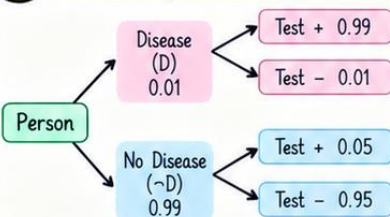
$$\begin{aligned} P(+) &= P(+ | D) \times P(D) + P(+ | \neg D) \times P(\neg D) \\ &= (0.99 \times 0.01) + (0.05 \times 0.99) \\ &= 0.0099 + 0.0495 = 0.0594 \end{aligned}$$

Now,

$$P(D | +) = \frac{0.99 \times 0.01}{0.0594} = \frac{0.0099}{0.0594} \approx 0.167$$

So, there is about a 16.7% chance of actually having the disease given a positive test result.

## 7 Tree Diagram View



Total probability of a positive test:  
 $P(+) = 0.99 \times 0.01 + 0.05 \times 0.99 = 0.0594$

## 8 Real-World Applications



Medical testing (disease diagnosis)



Spam detection (filtering emails)



Fraud detection (credit card transactions)



Machine learning (classification problems)



Risk assessment (finance, insurance)

## 9 Key Takeaways

- ✓ Bayes' Theorem helps update probabilities using new evidence.
- ✓  $P(A | B) = P(B | A) \times P(A) / P(B)$
- ✓ Prior, likelihood, evidence and posterior are key components.
- ✓ Useful in many real-world scenarios (medical, finance, AI, etc.).
- ✓ Even a highly accurate test can give surprising results (base rate fallacy).

"New evidence doesn't just give us answers, it gives us better questions."

# Random Variables

A random variable is a numerical value that represents the outcome of a random experiment.

Random variables turn uncertainty into numbers! 🎲

## 1 What is a Random Variable?

- A random variable (RV) assigns a numerical value to each outcome of a random experiment.
- It is a function that maps each outcome in the sample space to a real number.
- Usually denoted by a capital letter like  $X$ ,  $Y$ , or  $Z$ .
- Random variables can be discrete or continuous.

## 2 Types of Random Variables

### Discrete Random Variable

- Takes countable values (e.g. 0, 1, 2, ...).
- Described using a probability distribution (PMF).
- Example:**  $X$  = number of heads in 3 coin tosses  $\rightarrow X = \{0, 1, 2, 3\}$

### Continuous Random Variable

- Takes any value in a range (often infinitely many).
- Described using a probability density function (PDF).
- Example:**  $X$  = height of a person  $\rightarrow X$  can be any value (e.g. 165.2 cm).

## 3 Example: Discrete Random Variable

Let  $X$  = number of heads in 2 coin tosses.

Sample Space:  $S = \{HH, HT, TH, TT\}$

$X = \{0, 1, 2\}$



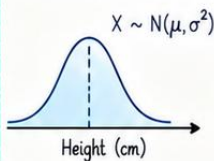
| Outcome | X (Number of Heads) | P(X)          |
|---------|---------------------|---------------|
| HH      | 2                   | $\frac{1}{4}$ |
| HT      | 1                   | $\frac{1}{4}$ |
| TH      | 1                   | $\frac{1}{4}$ |
| TT      | 0                   | $\frac{1}{4}$ |

This gives the Probability Mass Function (PMF) of  $X$ .

## 4 Example: Continuous Random Variable

Let  $X$  = height of a person (in cm).

- $X$  can take any value in a range (e.g. 150 to 200 cm).
- We describe  $X$  using a Probability Density Function (PDF), often assumed to be a normal distribution.



For a continuous RV,  $P(X = a) = 0$ . We work with probabilities over intervals, e.g.  $P(160 \leq X \leq 170)$ .

## 5 Probability Functions

### Probability Mass Function (PMF)

- Used for discrete random variables.
- gives the probability for each possible value.
- $P(X = x) \geq 0$
- $\sum P(X = x) = 1$

### Probability Density Function (PDF)

- Used for continuous random variables.
- $f(x) \geq 0$
- Total area under the curve = 1
- $P(a \leq X \leq b) = \int_a^b f(x) dx$

## 6 Expected Value and Variance

### Expected Value (Mean):

The long-run average value of a random variable.

- Discrete:  $E(X) = \sum x \cdot P(X = x)$
- Continuous:  $E(X) = \int_{-\infty}^{\infty} x \cdot f(x) dx$

### Variance:

Measures how much the values of a random variable vary around the mean.

- Discrete:  $\text{Var}(X) = \sum (x - \mu)^2 \cdot P(X = x)$
- Continuous:  $\text{Var}(X) = \int_{-\infty}^{\infty} (x - \mu)^2 \cdot f(x) dx$

Standard Deviation =  $\sqrt{\text{Var}(X)}$

## 7 Common Distributions (of Random Variables).

| Distribution | Type       | Parameters    | Example Use Case                        |
|--------------|------------|---------------|-----------------------------------------|
| Binomial     | Discrete   | $n, p$        | Number of successes in $n$ trials       |
| Poisson      | Discrete   | $\lambda$     | Number of events in a time interval     |
| Normal       | Continuous | $\mu, \sigma$ | Heights, test scores                    |
| Uniform      | Continuous | $a, b$        | Any value in $[a, b]$ is equally likely |
| Exponential  | Continuous | $\lambda$     | Time between events                     |

## 8 Key Takeaways

- ✓ A random variable assigns a numerical value to outcomes of a random experiment.
- ✓ Can be discrete or continuous.
- ✓ Discrete  $\rightarrow$  use PMF, Continuous  $\rightarrow$  use PDF.
- ✓ Expected value gives the long-run average.
- ✓ Variance measures the spread.
- ✓ Random variables are the foundation for many probability distributions and real-world applications.

## 9 Real-World Examples

- 🎲 Number of defects in a batch (discrete)
- 🕒 Time between customer arrivals (continuous)
- 📊 Daily stock returns (continuous)
- ✉ Number of emails received (discrete)
- 👥 Height of students in a class (continuous)

"A random variable is how we give a number to chance."

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# Expected Value ( $E[X]$ )

Expected value is the long-run average value of a random variable.  
It tells us the **center (mean)** of the probability distribution.

It's not what we get once, but what we expect on average in the long run! 😊

## 1 What is Expected Value?

- Expected value ( $E[X]$ ) is the **weighted average** of all possible values of a random variable, where the weights are the probabilities.
- It represents the **long-run average outcome** if we repeat the experiment many times.
- Also called the **mean** or **mathematical expectation**.

## 2 Formula (Discrete)

$$E[X] = \sum_{i=1}^n x_i \cdot P(X = x_i)$$

- $x_i$  = possible value of the random variable  $X$
- $P(X = x_i)$  = probability of  $x_i$
- Sum over all possible values.

## 3 Formula (Continuous)

$$E[X] = \int_{-\infty}^{\infty} x \cdot f(x) dx$$

- $f(x)$  = probability density function (PDF) of  $X$
- Integrate over all possible values.
- Used for continuous random variables (e.g. height, time, etc.).

## 4 Example 1: Fair Die

Let  $X$  = outcome when a fair six-sided die is rolled.  
Possible values: {1, 2, 3, 4, 5, 6}  
Each has probability  $1/6$ .



$$\begin{aligned} E[X] &= 1\left(\frac{1}{6}\right) + 2\left(\frac{1}{6}\right) + 3\left(\frac{1}{6}\right) + 4\left(\frac{1}{6}\right) + 5\left(\frac{1}{6}\right) + 6\left(\frac{1}{6}\right) \\ &= \frac{1}{6} (1 + 2 + 3 + 4 + 5 + 6) \\ &= \frac{1}{6} (21) = 3.5 \end{aligned}$$

On average, we expect 3.5 when rolling a fair die many times.

## 5 Example 2: Biased Coin

Let  $X = 1$  if Head, 0 if Tail.  
 $P(\text{Head}) = 0.7$ ,  $P(\text{Tail}) = 0.3$



$$\begin{aligned} E[X] &= 1(0.7) + 0(0.3) \\ &= 0.7 \end{aligned}$$

On average, we expect 0.7 heads per toss in the long run.

## 6 Example 3: Monthly Demand

A product can have demand of 0, 1, or 2 units with the following probabilities:

| X (Demand) | P(X) |
|------------|------|
| 0          | 0.2  |
| 1          | 0.5  |
| 2          | 0.3  |

$$\begin{aligned} E[X] &= 0(0.2) + 1(0.5) + 2(0.3) \\ &= 0 + 0.5 + 0.6 = 1.1 \end{aligned}$$

On average, we expect 1.1 units of demand per month.

## 7 Properties of Expected Value

- Linearity:**  $E[aX + b] = aE[X] + b$
- Constant:**  $E[c] = c$
- Sum:**  $E[X + Y] = E[X] + E[Y]$
- Scaling:**  $E[aX] = aE[X]$
- For independent variables:  
 $E[X \cdot Y] = E[X] \cdot E[Y]$

## 8 When is it Useful?

- ✓ Estimating long-term averages (e.g. returns, demand, cost).
- ✓ Decision making under uncertainty.
- ✓ Used in finance, insurance, ML, and many real-world applications.
- ✓ Helps compare different options based on average outcomes.



## 9 Key Takeaways

- ✓ Expected value is the long-run average of a random variable.
- ✓ For discrete variables, use a sum. For continuous variables, use an integral.
- ✓ It gives the "center" of the distribution but does not tell us about the spread (variance does that).
- ✓ Widely used in real-world problems like finance, risk analysis, and decision making.

"Expected value is not a promise, but an average over many possibilities."

# Probability Distributions

A probability distribution describes how the probability is spread across all possible values of a random variable.

Different distributions for different real-world situations! 😊

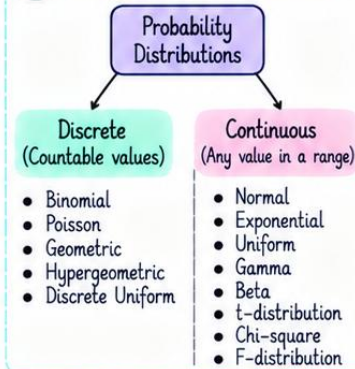
## 1 What is a Probability Distribution?

- A probability distribution shows the probability of each possible value of a random variable (RV).
- It can be **discrete** (countable values) or **continuous** (any value in a range).
- The total probability across all possible values is always 1.

$$\sum P(X = x) = 1 \quad (\text{discrete})$$

$$\int_{-\infty}^{\infty} f(x)dx = 1 \quad (\text{continuous})$$

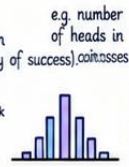
## 2 Types of Distributions



## 3 Common Discrete Distributions

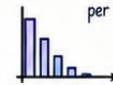
### Binomial Distribution

- Models number of successes in  $n$  independent trials ( $p$  = probability of success).
- $X \sim \text{Binomial}(n, p)$
- $P(X = k) = \binom{n}{k} p^k (1-p)^{n-k}$



### Poisson Distribution

- Models number of events in a fixed interval (e.g. time, space).
- $X \sim \text{Poisson}(\lambda)$
- $P(X = k) = \frac{e^{-\lambda} \lambda^k}{k!}$

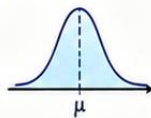


## 4 Common Continuous Distributions

### Normal Distribution

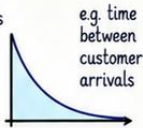
- Most important continuous distribution.
- Bell-shaped and symmetric.
- $X \sim N(\mu, \sigma^2)$

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$



### Exponential Distribution

- Models time between events (in a Poisson process).
- $X \sim \text{Exp}(\lambda)$
- $f(x) = \lambda e^{-\lambda x} \quad (x \geq 0)$



## 5 More Continuous Distributions

### Uniform Distribution

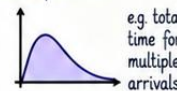
- All values in a range are equally likely.
- $X \sim U(a, b)$
- $f(x) = \frac{1}{b-a} \quad (a \leq x \leq b)$



### Gamma Distribution

- Models waiting time for multiple events.
- $X \sim \text{Gamma}(\alpha, \beta)$

$$f(x) = \frac{x^{\alpha-1} e^{-x/\beta}}{\beta^{\alpha} \Gamma(\alpha)}$$



## 6 Real-World Examples

| Distribution   | Example Use Case                           |
|----------------|--------------------------------------------|
| Binomial       | Number of defective items in a batch       |
| Poisson        | Customer arrivals at a store               |
| Geometric      | Number of trials until first success       |
| Hypergeometric | Drawing specific items without replacement |
| Normal         | Heights, exam scores, measurement errors   |
| Exponential    | Time between events (calls, failures)      |
| Uniform        | Random number generation                   |
| Gamma          | Total waiting time for multiple events     |
| Beta           | Modeling probabilities (e.g. success rate) |

## 7 Key Properties

### Discrete Distributions

- Defined using a Probability Mass Function (PMF).
- $P(X = x) \geq 0$
- $\sum P(X = x) = 1$
- Examples: Binomial, Poisson, Geometric.

### Continuous Distributions

- Defined using a Probability Density Function (PDF).
- $f(x) \geq 0$
- $\int_{-\infty}^{\infty} f(x) dx = 1$
- Probabilities are for ranges:  $P(a \leq X \leq b) = \int_a^b f(x) dx$
- Examples: Normal, Exponential.

## 8 Why are they Important?

- ✓ Help us model real-world uncertainty.
- ✓ Used in statistics, machine learning, finance, engineering, etc.
- ✓ Allow us to calculate probabilities, expected values and make decisions under uncertainty.
- ✓ Different distributions fit different types of data and problems.

## 9 Key Takeaways

- ✓ A probability distribution shows how probability is spread across values of a random variable.
- ✓ Discrete → use PMF, Continuous → use PDF.
- ✓ Choose the right distribution based on the real-world context.
- ✓ Distributions help us make sense of data and uncertainty.

"Probability distributions turn uncertainty into understanding."

 DataXOdyssey

# Binomial Distribution

A discrete probability distribution that models the number of successes in a fixed number of independent trials, each with the same probability of success.

Same experiment repeated  $n$  times, only two possible outcomes: success or failure! 😊

## 1 When to Use?

- ✓ Fixed number of trials ( $n$ ).
- ✓ Each trial has two outcomes (success or failure).
- ✓ Trials are independent.
- ✓ Probability of success ( $p$ ) is the same for all trials.
- ✓ We are interested in the number of successes ( $X$ ).

Examples:

- Number of defective items in a batch
- Number of heads in coin tosses
- Number of students who pass an exam
- Number of emails opened

## 2 Probability Mass Function (PMF)

If  $X \sim \text{Binomial}(n, p)$ , then

$$P(X = k) = \binom{n}{k} p^k (1-p)^{n-k}$$

where:

- $n$  = number of trials ( $n = 0, 1, 2, \dots$ ).
- $p$  = probability of success ( $0 \leq p \leq 1$ ).
- $k$  = number of successes ( $k = 0, 1, 2, \dots, n$ ).
- $\binom{n}{k} = \frac{n!}{k!(n-k)!}$

Note:  $0 \leq k \leq n$

$P(X = k) = 0$  for  $k < 0$  or  $k > n$

## 3 Mean and Variance

- Expected Value (Mean):

$$E[X] = n \cdot p$$

- Variance:

$$\text{Var}(X) = n \cdot p \cdot (1-p)$$

- Standard Deviation:

$$\sigma = \sqrt{n \cdot p \cdot (1-p)}$$

## 4 Example 1: Coin Tosses

A fair coin is tossed 10 times. What is the probability of getting exactly 4 heads?

Here,  $n = 10$ ,  $p = 0.5$ ,  $k = 4$

$$\begin{aligned} P(X = 4) &= \binom{10}{4} (0.5)^4 (0.5)^6 \\ &= 210 \times (0.5)^{10} \\ &= \frac{210}{1024} \approx 0.2051 \end{aligned}$$

There is about a 20.5% chance of getting exactly 4 heads in 10 tosses.



## 5 Example 2: Defective Items

In a batch of 20 items, the probability of a defective item is 0.1. What is the probability of getting exactly 3 defective items?

Here,  $n = 20$ ,  $p = 0.1$ ,  $k = 3$

$$\begin{aligned} P(X = 3) &= \binom{20}{3} (0.1)^3 (0.9)^{17} \\ &= 1140 \times (0.1)^3 \times (0.9)^{17} \\ &\approx 0.1368 \end{aligned}$$

There is about a 13.7% chance of getting exactly 3 defective items in the batch.



## 6 Probability Distribution Table

For a small  $n$ , we can list all probabilities.

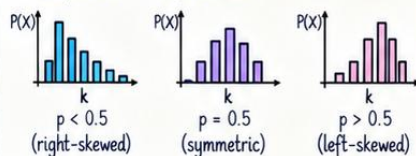
Example:  $X \sim \text{Binomial}(5, 0.4)$

| $k$ | $P(X = k)$ |
|-----|------------|
| 0   | 0.0778     |
| 1   | 0.2592     |
| 2   | 0.3456     |
| 3   | 0.2304     |
| 4   | 0.0768     |
| 5   | 0.0102     |

(Probabilities sum to 1)

## 7 Shape of the Distribution

- The shape depends on the value of  $p$ .
- If  $p = 0.5 \rightarrow$  symmetric (bell-shaped).
- If  $p < 0.5 \rightarrow$  right-skewed.
- If  $p > 0.5 \rightarrow$  left-skewed.



## 8 Key Takeaways

- ✓ Used for a fixed number of independent trials with two outcomes.
- ✓ Defined by parameters  $n$  and  $p$ .
- ✓  $P(X = k) = \binom{n}{k} p^k (1-p)^{n-k}$ .
- ✓ Mean =  $n \cdot p$ , Variance =  $n \cdot p \cdot (1-p)$ .
- ✓ Shape depends on  $p$ .
- ✓ Widely used in quality control, surveys, medicine, and many real-world problems.

## 9 Real-World Applications

- ⚙️ Manufacturing (Number of defective items)
- ✉️ Marketing (Number of emails opened)
- 👥 Education (Number of students passing)
- 📋 Healthcare (Number of patients responding to a treatment)

"Binomial distribution helps us model how often success happens in a world of repeated chances."

DataXOdyssey

# Poisson Distribution

A discrete probability distribution that models the number of events occurring in a fixed interval of time, space or area, when the events happen independently and at a constant average rate.

Not every important event is planned. Some just happen! 😊

## 1 When to Use?

- ✓ Counting the number of events in a fixed time, space or area.
- ✓ Events occur independently.
- ✓ Average rate ( $\lambda$ ) is constant.
- ✓ Events are rare (typically low probability in a small interval).
- ✓ Often used when  $n$  is large and  $p$  is small (Binomial approx.).

### Examples:

- Number of customer calls per hour
- Number of emails received per day
- Number of defects in a batch
- Number of accidents at an intersection
- Number of website visits per minute

## 2 Probability Mass Function (PMF)

If  $X \sim \text{Poisson}(\lambda)$ , then

$$P(X = k) = \frac{e^{-\lambda} \lambda^k}{k!}$$

$k = 0, 1, 2, \dots$

where:

- $\lambda$  (lambda) = average rate of events ( $\lambda > 0$ )
- $k$  = number of events (0, 1, 2, ...)
- $e \approx 2.718$  (base of natural logarithm)
- $k!$  =  $k$  factorial

Note:

- $P(X = k) \geq 0$  for all  $k$
- $\sum_{k=0}^{\infty} P(X = k) = 1$

## 3 Mean and Variance

- For a Poisson distribution:

$$E[X] = \lambda$$

$$\text{Var}(X) = \lambda$$

- The mean and variance are equal.
- Larger  $\lambda \rightarrow$  distribution is more spread out.

If  $\lambda$  is large (typically  $> 10$ ), the Poisson distribution can be approximated by a Normal distribution:

$$X \approx N(\lambda, \lambda)$$

As  $\lambda$  increases, the distribution becomes more symmetric.

## 4 Example 1: Website Visits

A website receives an average of 4 visits per minute. What is the probability of getting exactly 6 visits in a minute?

Here,  $\lambda = 4$ ,  $k = 6$

$$\begin{aligned} P(X = 6) &= \frac{e^{-4} 4^6}{6!} \\ &= \frac{0.0183 \times 4096}{720} \\ &\approx 0.1042 \end{aligned}$$



There is about a 10.42% chance of getting exactly 6 visits in a minute.

## 5 Example 2: Customer Calls

A call center receives an average of 3 calls per hour. What is the probability of receiving at least 5 calls in an hour?

Here,  $\lambda = 3$

$$\begin{aligned} P(X \geq 5) &= 1 - P(X \leq 4) \\ &= 1 - \sum_{k=0}^4 \frac{e^{-3} 3^k}{k!} \\ &= 1 - 0.8153 \\ &= 0.1847 \end{aligned}$$

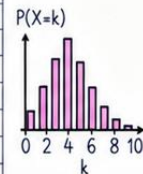


There is about an 18.47% chance of receiving at least 5 calls in an hour.

## 6 Poisson Distribution Table ( $\lambda = 3$ )

| k  | P(X = k) |
|----|----------|
| 0  | 0.0498   |
| 1  | 0.1494   |
| 2  | 0.2240   |
| 3  | 0.2240   |
| 4  | 0.1680   |
| 5  | 0.1008   |
| 6  | 0.0504   |
| 7  | 0.0216   |
| 8  | 0.0081   |
| 9  | 0.0027   |
| 10 | 0.0008   |

Probabilities sum to 1 (as  $k \rightarrow \infty$ ).



## 7 Properties

- Discrete distribution ( $k = 0, 1, 2, \dots$ ).
- Mean = Variance =  $\lambda$ .
- Models rare events.
- Arises as a limit of Binomial( $n, p$ ) when  $n$  is large and  $p$  is small ( $\lambda = np$ ).
- Sum of independent Poisson variables is also Poisson. If  $X \sim \text{Poi}(\lambda_1)$  and  $Y \sim \text{Poi}(\lambda_2)$ , then  $X + Y \sim \text{Poi}(\lambda_1 + \lambda_2)$ .

## 8 Real-World Applications

- Number of emails received per day
- Number of accidents at a junction
- Number of machine breakdowns in a factory
- Number of customers arriving at a store
- Number of network requests per second
- Number of rare side effects in clinical trials

## 9 Key Takeaways

- ✓ Poisson distribution models the number of events in a fixed interval.
- ✓  $P(X = k) = \frac{e^{-\lambda} \lambda^k}{k!}$
- ✓ Mean and variance are both  $\lambda$ .
- ✓ Useful for rare and independent events.
- ✓ Can approximate Binomial when  $n$  is large and  $p$  is small.
- ✓ Widely used in real-world problems across business, engineering, healthcare and technology.

"The Poisson distribution reminds us that even randomness has a pattern."

DataXOdyssey

# Normal Distribution

A continuous probability distribution that is symmetric and bell-shaped, commonly used to model real-world data due to the Central Limit Theorem.

Many real-world phenomena follow a normal distribution! 😊

## 1 When to Use?

- When data is **continuous**.
- Data is approximately **symmetric and bell-shaped**.
- Common in natural phenomena, measurement errors, test scores, heights, weights, etc.
- Often used due to the Central Limit Theorem (sum/mean of many independent variables  $\approx$  normal).

"If you are not sure, normal distribution is a good starting point!"

## 2 Probability Density Function (PDF)

If  $X \sim N(\mu, \sigma^2)$ , then the PDF is:

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

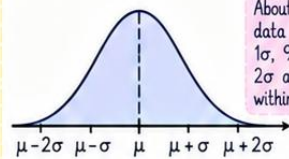
where:

- $\mu$  = mean (center of the distribution)
- $\sigma$  = standard deviation (spread)
- $\sigma^2$  = variance
- $x$  = any value of the random variable
- $e \approx 2.718$  (natural constant)
- $\pi \approx 3.1416$

The total area under the curve is 1  
 $\left( \int_{-\infty}^{\infty} f(x)dx = 1 \right)$

## 3 Shape of the Distribution

- Symmetric around the mean ( $\mu$ ).
- Bell-shaped curve.
- Mean = Median = Mode.
- Completely defined by two parameters:  $\mu$  and  $\sigma$ .
- Infinitely extends in both directions (but approaches 0).
- Points of inflection at  $\mu \pm \sigma$ .



About 68% of data lies within 1 $\sigma$ , 95% within 2 $\sigma$  and 99.7% within 3 $\sigma$ .

## 4 Standard Normal Distribution

- A normal distribution with:  $\mu = 0, \sigma = 1$
- Denoted as  $Z \sim N(0, 1)$ .
- We convert any normal variable  $X$  to a standard normal variable  $Z$  using:

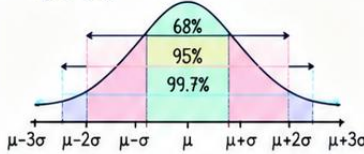
$$Z = \frac{X - \mu}{\sigma}$$

We then use the standard normal table (Z-table) to find probabilities.

## 5 Empirical Rule (68-95-99.7)

In a normal distribution:

- 68% of data lies within 1 standard deviation ( $\mu \pm 1\sigma$ )
- 95% within 2 standard deviations ( $\mu \pm 2\sigma$ )
- 99.7% within 3 standard deviations ( $\mu \pm 3\sigma$ )



## 6 Example: Test Scores

Test scores in a class are normally distributed with  $\mu = 70$  and  $\sigma = 10$ . What is the probability that a student scores more than 85?

Step 1: Convert to Z-score

$$Z = \frac{85 - 70}{10} = 1.5$$

Step 2: Use Z-table

$$\begin{aligned} P(Z > 1.5) &= 1 - P(Z \leq 1.5) \\ &= 1 - 0.9332 \\ &= 0.0668 \end{aligned}$$

There is about a 6.68% chance that a student scores more than 85.



## 7 Common Applications

- Heights and weights (human characteristics)
- Measurement errors (instruments and processes)
- Test scores (education)
- Financial returns (often assumed normal)
- Natural phenomena (e.g. blood pressure, IQ scores, etc.)

## 8 Key Takeaways

- ☒ Normal distribution is continuous, symmetric and bell-shaped.
- ☒ Defined by mean ( $\mu$ ) and standard deviation ( $\sigma$ ).
- ☒ Use Z-score to standardize values.
- ☒ Empirical rule helps us quickly understand data spread.
- ☒ Widely used in real-world problems due to the Central Limit Theorem.
- ☒ Always check if the data is approximately normal before using it.

## 9 Useful Formulas

- PDF:  $f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$
- Standardization:  $Z = \frac{X - \mu}{\sigma}$
- Mean:  $E[X] = \mu$
- Variance:  $\text{Var}(X) = \sigma^2$
- Empirical Rule:  $P(\mu \pm 1\sigma) \approx 68\%$   
 $P(\mu \pm 2\sigma) \approx 95\%$   
 $P(\mu \pm 3\sigma) \approx 99.7\%$

"The normal distribution shows up everywhere in the real world."

DataXOdyssey

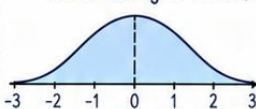
# Standard Normal Distribution & Z-Score

The standard normal distribution is a normal distribution with mean 0 and standard deviation 1. The Z-score tells us how many standard deviations a value is from the mean.

Z-scores turn any normal distribution into the same standard scale! 😊

## 1 What is Standard Normal Distribution?

- A normal distribution with mean ( $\mu$ ) = 0 and standard deviation ( $\sigma$ ) = 1.
- Denoted as  $Z \sim N(0, 1)$ .
- It is symmetric, bell-shaped and has total area 1.
- Used as a reference to find probabilities for any normal distribution by standardizing values (using Z-scores).



Total area under the curve = 1

## 2 Z-Score

- A Z-score tells us how many standard deviations a value  $X$  is from the mean.
- Formula:

$$Z = \frac{X - \mu}{\sigma}$$

- $X$  = value
- $\mu$  = mean
- $\sigma$  = standard deviation
- $Z$  = standard score

$Z > 0 \rightarrow$  value is above the mean

$Z < 0 \rightarrow$  value is below the mean

## 3 Properties

- Mean = 0
- Standard deviation = 1
- Symmetric around 0
- Total area under the curve = 1
- 50% of the data lies below 0 and 50% above 0
- Probabilities depend only on the Z-value (not on the original units)

## 4 Converting to Z-Score (Example)

A test score is 82, with a class mean of 70 and a standard deviation of 6. Find the Z-score.

$$Z = \frac{X - \mu}{\sigma} = \frac{82 - 70}{6}$$

$$Z = \frac{12}{6} = 2$$

Same formula works for any normal distribution!

The score is 2 standard deviations above the mean.

## 5 Finding Probabilities (Examples)

a)  $P(Z < 1.5)$

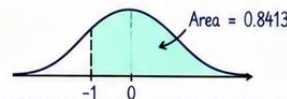
From Z-table:  $P(Z < 1.5) = 0.9332$



b)  $P(Z > -1)$

From Z-table:  $P(Z < -1) = 0.1587$

$P(Z > -1) = 1 - 0.1587 = 0.8413$



## 6 Using the Z-Table

Step 1: Find the row for the first decimal place (e.g. 1.5  $\rightarrow$  row 1.5).

Step 2: Find the column for the second decimal place (e.g. 1.53  $\rightarrow$  col 0.03).

Step 3: The value gives  $P(Z < z)$ .

Z-Table (excerpt)

| z   | 0.00   | 0.01   | 0.02   | 0.03   |
|-----|--------|--------|--------|--------|
| 1.5 | 0.9332 | 0.9345 | 0.9357 | 0.9370 |
| 1.6 | 0.9452 | 0.9463 | 0.9474 | 0.9484 |

For negative Z, use symmetry:

$$P(Z < -z) = 1 - P(Z < z)$$

## 7 Common Z-Values

| Z  | $P(Z < z)$ |
|----|------------|
| -2 | 0.0228     |
| -1 | 0.1587     |
| 0  | 0.5000     |
| 1  | 0.8413     |
| 2  | 0.9772     |

Remember: 68 - 95 - 99.7 rule also applies here (same as normal distribution).

## 8 Real-World Applications

- Standardizing test scores
- Comparing data from different distributions
- Quality control (e.g. product measurements)
- Finance (e.g. returns, risk analysis)
- Healthcare (e.g. patient measurements)
- Machine learning (feature scaling)

## 9 Key Takeaways

- Standard normal distribution:  $Z \sim N(0, 1)$
- Z-score:  $Z = (X - \mu) / \sigma$
- Z tells us how many standard deviations a value is from the mean
- Use Z-table to find probabilities
- Same approach works for any normal distribution
- Widely used in statistics, data analysis and real-world problems

"A Z-score puts everyone on the same scale."

DataXOdyssey

# Central Limit Theorem

The Central Limit Theorem (CLT) states that, for a large enough sample size, the sampling distribution of the mean ( $\bar{x}$ ) will be approximately normal, regardless of the shape of the original population distribution.

No matter what the population looks like... the sample mean becomes normal! 😊

## 1 When Does It Apply?

- Sample size ( $n$ ) is large enough (Generally,  $n \geq 30$  is used as a rule of thumb).
- Observations are independent (from each other).
- Population has a finite variance ( $\sigma^2$  is not infinite).

It works for any population shape — normal, uniform, skewed, etc.

## 2 What Does It Say?

If  $X_1, X_2, \dots, X_n$  is a random sample from any population with mean  $\mu$  and standard deviation  $\sigma$ , then the sampling distribution of  $\bar{x}$  is:  $\bar{x} \sim N(\mu, \frac{\sigma}{\sqrt{n}})$

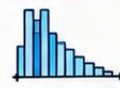
$$\bar{x} \sim N\left(\mu, \frac{\sigma}{\sqrt{n}}\right)$$

where:

- $\mu$  = population mean
- $\sigma$  = population standard deviation
- $n$  = sample size

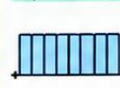
## 3 Visual Example

Population (skewed)



Sample mean (large  $n$ )  $\sim$  Normal

Population (uniform)



Sample mean (large  $n$ )  $\sim$  Normal

Population (bimodal)



Sample mean (large  $n$ )  $\sim$  Normal

## 4 Example

Scores of 50 students in a class are normally distributed with mean 70 and standard deviation 10.

What is the distribution of the sample mean ( $n = 50$ )?

$$\begin{aligned}\bar{x} &\sim N\left(70, \frac{10}{\sqrt{50}}\right) \\ &= N(70, 1.41)\end{aligned}$$

Even if the original scores were skewed, with  $n = 50$ , the sample mean will be approximately normal.

## 5 Key Takeaways

- ✓ Applies when sample size is large (usually  $n \geq 30$ ).
- ✓ Sample mean is normally distributed (approx.).
- ✓ Mean of the sampling distribution = population mean ( $\mu$ ).
- ✓ Standard deviation of sampling distribution =  $\frac{\sigma}{\sqrt{n}}$ .
- ✓ Works for any population shape (as long as variance is finite).

## 6 Why Is It Important?



Helps us make inferences about the population using sample data.



Forms the basis for many statistical methods (confidence intervals, hypothesis testing, etc.).



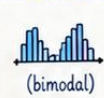
Makes analysis easier, even when the population distribution is unknown or non-normal.



Widely used in real-world fields like finance, healthcare, quality control, and more.

## 7 Quick Recap

Any population (distribution shape)



Take a large sample ( $n \geq 30$ )

Sampling distribution of  $\bar{x}$  is approximately normal

$$\bar{x} \sim N\left(\mu, \frac{\sigma}{\sqrt{n}}\right)$$

## 8 Remember

"The Central Limit Theorem is like a magic wand... no matter how messy the population, the sample mean behaves nicely!" 😊



"Larger samples, simpler stories."

 DataXOdyssey

# Sampling Distribution

A sampling distribution shows the probability distribution of a statistic (like the sample mean) calculated from all possible samples of a fixed size drawn from a population.

Different samples, same population, new distribution! 😊

## 1 What is a Sampling Distribution?

- It is the distribution of a sample statistic (e.g., mean, proportion) obtained from all possible samples of a given size from a population.
- Each sample gives a different statistic, so these statistics form a distribution.
- Helps us understand how much the statistic varies from sample to sample.

The population is fixed, but the samples (and their statistics) keep changing!

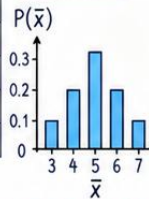
## 2 Example: Sampling Distribution of the Mean

Suppose a population has values: 2, 4, 6, 8

Take all possible samples of size  $n = 2$  (without replacement).

| Sample | Sample Mean ( $\bar{x}$ ) |
|--------|---------------------------|
| (2, 4) | 3                         |
| (2, 6) | 4                         |
| (2, 8) | 5                         |
| (4, 6) | 5                         |
| (4, 8) | 6                         |
| (6, 8) | 7                         |

Sampling distribution of  $\bar{x}$ :



## 3 Key Properties

- It is a probability distribution of a statistic (not raw data).
- Depends on the population distribution and sample size ( $n$ ).
- As sample size increases, the sampling distribution of the mean becomes approximately normal (Central Limit Theorem).
- The mean of the sampling distribution equals the population parameter.
- The variability of the sampling distribution is measured by the standard error.

## 4 Mean and Standard Error

If the population has mean  $\mu$  and standard deviation  $\sigma$ , then for the sample mean  $\bar{x}$  (with sample size  $n$ ):

Mean of sampling distribution:

$$E(\bar{x}) = \mu$$

Standard error (of the mean):

$$SE(\bar{x}) = \frac{\sigma}{\sqrt{n}}$$

Larger sample size  $\rightarrow$  smaller standard error  $\rightarrow$  less variability in sample means.

## 5 Shape of the Sampling Distribution

Population Normal



Sampling distribution of  $\bar{x}$  is normal (for any  $n$ )

Population Non-Normal (skewed)



Sampling distribution of  $\bar{x}$  becomes more normal as  $n$  increases

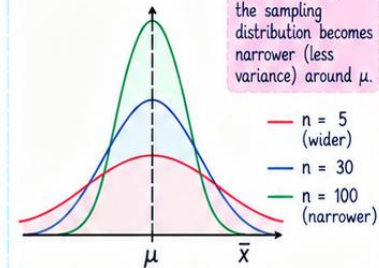
Population Uniform



Sampling distribution of  $\bar{x}$  is approximately normal for large  $n$

## 6 Effect of Sample Size

As  $n$  increases, the sampling distribution becomes narrower (less variance) around  $\mu$ .



## 7 Sampling Distribution of the Proportion

For a population proportion  $p$ , the sampling distribution of  $\hat{p}$  (sample proportion) has:

Mean:  $E(\hat{p}) = p$

Standard error:  $SE(\hat{p}) = \sqrt{\frac{p(1-p)}{n}}$

Conditions:  $n$  is large enough (typically  $np \geq 10$  and  $n(1-p) \geq 10$ ).

## 8 Why is it Important?



Helps us make inferences about population parameters using sample data.



Used in confidence intervals and hypothesis testing.



Quantifies the uncertainty in sample estimates.



Foundation for many statistical methods.

## 9 Key Takeaways

- A sampling distribution is the distribution of a statistic from all possible samples of a fixed size.
- Mean of sampling distribution = population parameter.
- Standard error measures variability.
- Shape depends on population distribution and sample size.
- For large  $n$ , the sampling distribution of the mean is approximately normal (Central Limit Theorem).
- Essential for inference in statistics.

"Samples change. Patterns emerge."

 DataXOdyssey

# Standard Error

The standard error (SE) measures how much a sample statistic (like the mean) is likely to vary from the true population parameter due to random sampling.

Smaller standard error  
= More confident in our estimate! 😊

## 1 What is Standard Error?

- Standard error is the standard deviation of a sampling distribution.
- It tells us how much a sample statistic is expected to vary across different samples (of the same size) from the same population.
- It is not the same as standard deviation ( $\sigma$ ), which measures variability in the data itself.

"Standard error measures the accuracy of an estimate."

## 2 Formula (Mean)

The standard error of the sample mean ( $\bar{x}$ ) is:

$$SE(\bar{x}) = \frac{\sigma}{\sqrt{n}}$$

where:

- $\sigma$  = population standard deviation
- $n$  = sample size
- $\bar{x}$  = sample mean

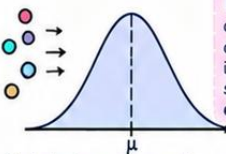
If population standard deviation ( $\sigma$ ) is unknown, we use the sample standard deviation ( $s$ ):

$$SE(\bar{x}) = \frac{s}{\sqrt{n}}$$

## 3 Intuition

- If we take many samples from the same population, each sample will have a different mean.
- These sample means form a sampling distribution.
- The standard error is the standard deviation of that distribution.

Multiple samples ( $n = 30$ )



The spread of this distribution is the standard error.

Distribution of sample means

## 4 Example

A population has  $\sigma = 20$ . We take samples of size  $n = 25$ . What is the standard error of the sample mean?

$$SE(\bar{x}) = \frac{20}{\sqrt{25}} = \frac{20}{5} = 4$$

So the standard deviation of the sampling distribution of the mean is 4.

## 5 Effect of Sample Size

- As sample size ( $n$ ) increases, the standard error decreases.
- Larger samples give more precise estimates.
- The relationship is inversely proportional to the square root of  $n$ .

Double the sample size  
→ standard error reduces by  $1/\sqrt{2}$  ( $\approx 0.71$  times).

## 6 Standard Error vs Standard Deviation

| Standard Deviation ( $\sigma$ )                 | Standard Error (SE)                                             |
|-------------------------------------------------|-----------------------------------------------------------------|
| Measures variability in individual data points. | Measures variability in a sample statistic (e.g., sample mean). |
| Describes the spread of the data.               | Describes the accuracy (precision) of an estimate.              |
| Does not depend on sample size.                 | Decreases as sample size increases ( $SE \propto 1/\sqrt{n}$ ). |

## 7 Standard Error for Other Statistics

- For a sample proportion ( $\hat{p}$ ):

$$SE(\hat{p}) = \sqrt{\frac{p(1-p)}{n}}$$

- For a sample sum:

$$SE(S) = \sigma \sqrt{n}$$

## 8 Key Takeaways

- ✓ Standard error measures the variability of a sample statistic.
- ✓  $SE(\bar{x}) = \frac{\sigma}{\sqrt{n}}$  (or  $\frac{s}{\sqrt{n}}$  if  $\sigma$  unknown).
- ✓ Larger sample size → smaller standard error.
- ✓ Used in confidence intervals, hypothesis testing, and many statistical methods.
- ✓ Smaller standard error means a more precise estimate.

## 9 Real-World Applications



Market research  
(e.g., estimating customer satisfaction)



Surveys and opinion polls  
(e.g., election polling)



Clinical trials  
(e.g., estimating treatment effects)



Business analytics  
(e.g., average revenue, conversion rates)

"More data, less uncertainty."

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# Confidence Intervals

A confidence interval gives a range of plausible values for a population parameter based on sample data. It gives us a level of confidence that the true parameter lies in this range.

It's a range, not a single number! We're estimating, not guessing! 😊

## 1 What is a Confidence Interval?

- A confidence interval (CI) is a range of values, calculated from sample data, that is likely to contain the true population parameter (e.g., mean  $\mu$  or proportion  $p$ ).
- It is based on a confidence level (e.g., 90%, 95%, 99%).
- It accounts for sampling variability using the standard error.

Repeated sampling would produce different intervals, and a certain percentage of them will contain the true parameter.

## 2 Confidence Interval for a Mean

When the population standard deviation ( $\sigma$ ) is unknown (common in practice), we use the t-distribution:

$$CI = \bar{x} \pm t_{\alpha/2, n-1} \times SE(\bar{x})$$

where:

- $\bar{x}$  = sample mean
- $t_{\alpha/2, n-1}$  = critical value from t-table
- $SE(\bar{x}) = \frac{s}{\sqrt{n}}$  (standard error of the mean)
- $n$  = sample size
- $\alpha = 1 - \text{confidence level}$

If population standard deviation ( $\sigma$ ) is known, use Z instead of t.

## 3 Confidence Interval for a Proportion

For a population proportion ( $p$ ):

$$CI = \hat{p} \pm z_{\alpha/2} \times \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$$

where:

- $\hat{p}$  = sample proportion
- $z_{\alpha/2}$  = critical value from standard normal distribution (Z-table)
- $n$  = sample size

Conditions:  $n$  is large enough (typically  $np \geq 10$  and  $n(1-p) \geq 10$ ).

## 4 Example (Mean)

A sample of 36 students has a mean score of 78 and a standard deviation of 12. Find the 95% confidence interval for the population mean.

$$\begin{aligned} CI &= 78 \pm t_{0.025, 35} \times \frac{12}{\sqrt{36}} \\ &= 78 \pm 2.03 \times 2 \\ &= 78 \pm 4.06 \\ &= (73.94, 82.06) \end{aligned}$$

We are 95% confident that the true mean score lies between 73.94 and 82.06.

## 5 Example (Proportion)

In a survey, 200 out of 500 customers prefer a product. Find the 95% confidence interval for the population proportion.

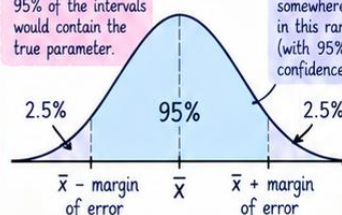
$$\begin{aligned} \hat{p} &= \frac{200}{500} = 0.4 \\ CI &= 0.4 \pm 1.96 \times \sqrt{\frac{0.4(1-0.4)}{500}} \\ &= 0.4 \pm 1.96 \times 0.0219 \\ &= 0.4 \pm 0.0429 \\ &= (0.357, 0.443) \end{aligned}$$

We are 95% confident that the true proportion lies between 35.7% and 44.3%.

## 6 Visualizing a 95% Confidence Interval

If we took many samples, about 95% of the intervals would contain the true parameter.

The true parameter is somewhere in this range (with 95% confidence).



## 7 Common Confidence Levels

| Confidence Level | Critical Value |
|------------------|----------------|
| 90%              | 1.645          |
| 95%              | 1.96           |
| 99%              | 2.576          |

Higher confidence level → wider interval (because the critical value is larger).

## 8 Key Takeaways

- ✓ A confidence interval gives a range, not a single estimate.
- ✓ It accounts for sampling variability using the standard error.
- ✓ Wider intervals for higher confidence levels.
- ✓ Larger sample size → narrower intervals.
- ✓ Used widely in surveys, experiments and real-world decision making.

## 9 Real-World Applications

- Market research (e.g., customer satisfaction)
- Healthcare (e.g., treatment effectiveness)
- A/B testing (e.g., conversion rates)
- Business analytics (e.g., average revenue, churn rate)
- Policy making & social studies (e.g., population estimates)

"Not a point, but a range of possibilities."

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# Hypothesis Testing

Hypothesis testing is a statistical method to decide, based on sample data, whether there is enough evidence to reject a claim about a population parameter.

We don't prove something true. We just find enough evidence to question it! 😊

## 1 What is Hypothesis Testing?

- A method to make a decision about a population parameter using sample data.
- We start with a claim (null hypothesis) and check if the sample data provides enough evidence to reject it.
- It helps us determine if an observed effect is due to chance or is statistically significant.

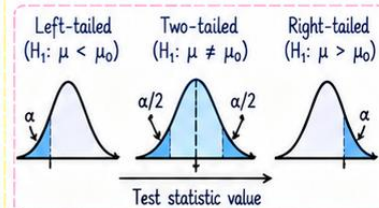
It's not about proving something is true, but about finding evidence to reject the null hypothesis.

## 2 Key Terms

- Null Hypothesis ( $H_0$ ):** Default claim (usually no effect or no difference).
- Alternative Hypothesis ( $H_1$  or  $H_a$ ):** Opposite of  $H_0$  (what we want to find evidence for).
- Test Statistic:** A value calculated from sample data (e.g.,  $z$ ,  $t$ ,  $\chi^2$ ).
- p-value:** Probability of observing the data (or more extreme) if  $H_0$  is true.
- Significance Level ( $\alpha$ ):** Threshold to decide (commonly 0.05).
- Critical Region:** Values of test statistic that lead to rejection of  $H_0$ .

## 3 Types of Hypothesis Tests

- One-tailed test:** Tests for an effect in one direction ( $>$  or  $<$ ).
- Two-tailed test:** Tests for an effect in both directions ( $\neq$ ).



## 4 Steps in Hypothesis Testing

- State the null and alternative hypotheses ( $H_0$  and  $H_1$ ).
- Choose the significance level ( $\alpha$ ).
- Select the appropriate test statistic ( $z$ ,  $t$ ,  $\chi^2$ , etc.).
- Calculate the test statistic and p-value.
- Make a decision (reject or fail to reject  $H_0$ ).
- Draw a conclusion in context.

## 5 Example (One-Sample Z-Test)

A company claims the average delivery time is 2 days. A sample of 36 orders has a mean delivery time of 2.5 days with a known population standard deviation of 1 day. Test at  $\alpha = 0.05$  if the average delivery time is higher than 2 days.

$$H_0: \mu = 2 \quad H_1: \mu > 2$$

Test statistic:

$$z = \frac{\bar{x} - \mu_0}{\frac{\sigma}{\sqrt{n}}} = \frac{2.5 - 2}{\frac{1}{\sqrt{36}}} = \frac{0.5}{0.1667} = 3.00$$

$$p\text{-value} = P(Z > 3.00) = 0.0013$$

Since  $p\text{-value} (0.0013) < 0.05$ , we reject  $H_0$ . There is sufficient evidence that the average delivery time is higher than 2 days.

## 6 p-value and Decision Rule

- If  $p\text{-value} \leq \alpha \rightarrow$  Reject  $H_0$  (statistically significant)
- If  $p\text{-value} > \alpha \rightarrow$  Fail to reject  $H_0$  (not enough evidence)

A small p-value means the observed result is unlikely under  $H_0$ .

Decision Rule ( $\alpha = 0.05$ )

| p-value     | Decision             |
|-------------|----------------------|
| $\leq 0.05$ | Reject $H_0$         |
| $> 0.05$    | Fail to reject $H_0$ |

## 7 Common Tests

| Test            | Use Case                                         | Test Statistic |
|-----------------|--------------------------------------------------|----------------|
| Z-test          | Mean ( $\sigma$ known or large $n$ )             | $Z$            |
| t-test          | Mean ( $\sigma$ unknown, small $n$ )             | $t$            |
| Chi-square test | Categorical data (goodness of fit, independence) | $\chi^2$       |
| F-test          | Compare variances                                | $F$            |
| Proportion test | Population proportion                            | $Z$            |

## 8 Key Takeaways

- Start with a null hypothesis.
- Use appropriate test and check assumptions.
- p-value helps in decision making.
- Rejecting  $H_0$  means evidence against the claim, not proof.
- Results should always be interpreted in context.
- Statistical significance  $\neq$  practical significance.

## 9 Real-World Applications

- A/B testing (e.g., website conversion rates)
- Clinical trials (e.g., new drug effectiveness)
- Market research (e.g., customer preferences)
- Quality control (e.g., defect rate)
- Finance (e.g., comparing investment returns)

"Data speaks. Hypothesis testing helps us listen."

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# P-Value & Statistical Significance

The p-value tells us how likely we would observe our data (or something more extreme) if the null hypothesis were true. It helps us decide whether the result is statistically significant.

A small p-value doesn't prove a claim, it just gives strong evidence! 😊

## 1 What is a p-value?

- The p-value is the probability of observing the sample data (or more extreme) if the null hypothesis ( $H_0$ ) is true.
- It measures how compatible the data are with  $H_0$ .
- A smaller p-value means the data are unlikely under  $H_0$ , providing evidence against  $H_0$ .

"The p-value assumes the null hypothesis is true."

## 2 How is the p-value Calculated?

- First, choose a test statistic (e.g.,  $z$ ,  $t$ ,  $\chi^2$ ) based on the problem.
- Calculate the test statistic from your sample data.
- Find the probability of observing a value as extreme (or more extreme) than the test statistic under the distribution assumed by  $H_0$ .

$p\text{-value} = P(\text{Test statistic} \geq \text{observed})$   
(or  $\leq$  observed, depending on the test)

## 3 Interpreting the p-value

| p-value range        | Interpretation                      |
|----------------------|-------------------------------------|
| $p \leq 0.01$        | Very strong evidence against $H_0$  |
| $0.01 < p \leq 0.05$ | Strong evidence against $H_0$       |
| $0.05 < p \leq 0.10$ | Moderate evidence against $H_0$     |
| $p > 0.10$           | Not enough evidence to reject $H_0$ |

Smaller p-value  $\rightarrow$  stronger evidence against the null hypothesis.

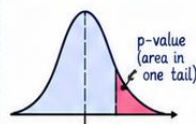
## 4 Relation to Significance Level ( $\alpha$ )

- The significance level ( $\alpha$ ) is a predefined threshold (commonly 0.05).
- Compare the p-value with  $\alpha$ :
- If  $p\text{-value} \leq \alpha \rightarrow$  Reject  $H_0$  (result is statistically significant).
- If  $p\text{-value} > \alpha \rightarrow$  Fail to reject  $H_0$  (not enough evidence).

$\alpha$  is chosen before looking at the data.

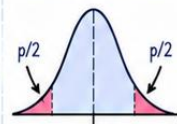
## 5 One-tailed vs Two-tailed p-value

One-tailed test  
(testing for  $>$  or  $<$ )



$H_1: \mu > \mu_0$  (right-tailed)  
 $H_1: \mu < \mu_0$  (left-tailed)

Two-tailed test  
(testing for  $\neq$ )



$H_1: \mu \neq \mu_0$   
p-value = area in both tails  
(2 x one tail)

## 6 Example

A company claims the average delivery time is 2 days. A sample of 36 orders has a mean of 2.5 days with a known  $\sigma = 1$  day. Test at  $\alpha = 0.05$ .

$$H_0: \mu = 2 \quad H_1: \mu > 2$$

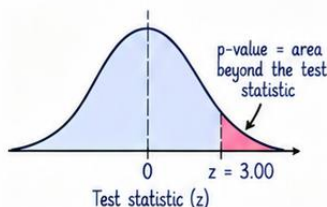
Test statistic ( $z$ ):

$$z = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}} = \frac{2.5 - 2}{1/\sqrt{36}} = \frac{0.5}{0.1667} = 3.00$$

$$p\text{-value} = P(Z > 3.00) = 0.0013$$

Since p-value (0.0013)  $<$  0.05, we reject  $H_0$ . The result is statistically significant.

## 7 Visualizing p-value



The shaded area shows how extreme our data is under  $H_0$ .

## 8 Important Things to Remember

- ✓ A small p-value does NOT prove the alternative hypothesis, it only suggests evidence against  $H_0$ .
- ✓ A large p-value does NOT prove  $H_0$ , it only means insufficient evidence.
- ✓ The p-value depends on sample size (larger samples can give small p-values for small effects).
- ✓ Statistical significance is not the same as practical significance (consider effect size and context).
- ✓ Always decide  $\alpha$  before looking at the data.

## 9 Real-World Applications

- A/B testing (e.g., website conversion rates)
- Clinical trials (e.g., new drug effectiveness)
- Market research (e.g., customer preferences)
- Quality control (e.g., defect rates)
- Business analytics (e.g., impact of a new feature)
- Social sciences (e.g., survey analysis)

"A small p-value speaks loudly, but context speaks louder."

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# Type I & Type II Errors

In hypothesis testing, we might make the wrong decision. There are two types of errors: Type I error and Type II error.

We don't always get it right! Statistics helps us manage the risk of being wrong. 😊

## 1 What are Type I & Type II Errors?

- In hypothesis testing, we make a decision to either reject or fail to reject the null hypothesis ( $H_0$ ).
- Sometimes, this decision is wrong.
- Type I and Type II errors describe the two possible wrong decisions we can make.
- We can't eliminate both errors, but we can control the probability of Type I error ( $\alpha$ ).

"In real life, there is uncertainty, and we might be wrong."

## 2 The Two Types of Errors

### Type I Error (False Positive)

- Rejecting a true null hypothesis.
- Concluding that an effect exists when in reality, it does not.
- Probability =  $\alpha$  (significance level, e.g., 0.05).

Example: Concluding a new drug works when it actually doesn't.

### Type II Error (False Negative)

- Failing to reject a false null hypothesis.
- Concluding that there is no effect when in reality, an effect exists.
- Probability =  $\beta$ .

Example: Concluding a new drug doesn't work when it actually does.

## 3 Decision Outcomes

| True Situation                 | Reject $H_0$                                                          | Fail to Reject $H_0$                                           |
|--------------------------------|-----------------------------------------------------------------------|----------------------------------------------------------------|
| $H_0$ is true (No effect)      | Type I Error (False Positive)<br>Probability = $\alpha$               | Correct Decision (True Negative)<br>Probability = $1 - \alpha$ |
| $H_0$ is false (Effect exists) | Correct Decision (True Positive)<br>Probability = $1 - \beta$ (Power) | Type II Error (False Negative)<br>Probability = $\beta$        |

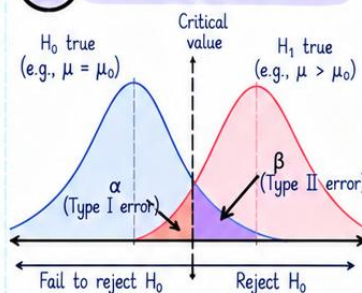
Type I error → Seeing something that isn't there.  
Type II error → Missing something that is there.

## 4 Probabilities and Notation

- $\alpha$  (alpha) = P(Type I error) = significance level (commonly 0.05).
- $\beta$  (beta) = P(Type II error).
- $1 - \beta$  = Power of the test (probability of correctly rejecting  $H_0$  when it is false).

We usually choose a small  $\alpha$  (e.g., 0.05), but  $\beta$  depends on the effect size, sample size and variability.

## 5 Visualizing the Errors



## 6 Trade-off Between $\alpha$ and $\beta$

- If we decrease  $\alpha$  (make it harder to reject  $H_0$ ), the probability of Type I error decreases.
- But this usually increases  $\beta$  (makes it harder to detect a true effect).
- To reduce  $\beta$ , we can:
  - Increase the sample size ( $n$ ).
  - Increase the effect size (larger difference).
  - Reduce variability in the data.
  - Use a more powerful test.

There is a trade-off: we cannot make both  $\alpha$  and  $\beta$  very small at the same time.

## 7 Example (Medical Testing)

- A new test is designed to detect a disease.
- Type I error: Test says the person has the disease, but they actually don't (false alarm).
  - Type II error: Test says the person doesn't have the disease, but they actually do (missed case).

In medical tests, both errors have real consequences, so the balance between  $\alpha$  and  $\beta$  is important.

## 8 Key Takeaways

- ✓ Type I error = reject a true  $H_0$  (probability =  $\alpha$ ).
- ✓ Type II error = fail to reject a false  $H_0$  (probability =  $\beta$ ).
- ✓ Power =  $1 - \beta$ .
- ✓ We usually set  $\alpha = 0.05$ .
- ✓ Reducing  $\alpha$  often increases  $\beta$  (trade-off).
- ✓ Larger sample size and effect size help reduce  $\beta$ .
- ✓ The choice of  $\alpha$  and  $\beta$  depends on the context and consequences.

## 9 Real-World Applications

- Clinical trials (does a new drug work?)
- A/B testing (is a new feature better?)
- Quality control (is a product defective?)
- Policy decisions (is a program effective?)
- Fraud detection (is a transaction suspicious?)

"Every decision has a risk — understand the errors."

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